1 **(a)** $2^3 \times 3^2 \times 7$ **[B2]**

M1 for a correct partial factorisation (e.g., 8×63 or 2×252) or a correct prime factor tree/ladder.

(b) 14 **[B2]**

M1 for identifying the needed powers 2×7 or writing $2^4 \times 3^2 \times 7^2$.

[Total Marks: 4]

2 **(a)** $2^2 \times 3 \times 7$ **[B2]**

M1 for $2, 2, 3, 7$ correctly identified in a factor tree or division ladder

(b) 252 **[B2]**

M1 for writing $126 = 2 \times 3^2 \times 7$ or writing multiples of 84 and 126

[Total Marks: 4]

3 **(a)** **(i)** 12 or 36 **[B1]**

(ii) 64 **[B1]**

(iii) 23 **[B1]**

(b) Prime factors: $36 = 2^2 \times 3^2$ and $64 = 2^6$ **[M1]**

LCM $= 2^6 \times 3^2 = 576$ **[A1]**

[Total Marks: 5]

4 **(a)** 13 **[B1]**

(b) $\sqrt{7}$ **[B1]**

(c) -5 **[B1]**

(d) 0.25 **[B1]**

(e) 64 **[B1]**

[Total Marks: 5]

5 **(a)** **(i)** 23 **[B1]**

(ii) 8 or 27 or 64 **[B1]**

(iii) 64 **[B1]**

(b) $2^2 \times 3^2$ (or $2 \times 2 \times 3 \times 3$) **[B2]**

Partial working: Correct factor tree or repeated division showing 2 and 3 as prime factors **[M1]**

[Total Marks: 5]

6 **(a)** **(i)** 2 or 23 **[B1]**

(ii) 27 or 64 **[B1]**

(iii) 64 **[B1]**

(iv) 15 **[B1]**

(b) $2^3 \times 3^2$ **[B2]**

If 0 scored, M1 for $2 \times 2 \times 2 \times 3 \times 3$ or a correct factor tree/division **[M1]**

[Total Marks: 6]

7 **(a)** Factor tree or repeated division showing prime factors 2, 2, 2, 3, 7 **[M1]**

$2^3 \times 3 \times 7$ **[A1]**

(b) $2^2 \times 3 \times 7$ or matching list of factors **[M1]**

84 **[A1]**

(c) $2^3 \times 3^2 \times 7$ or matching list of multiples **[M1]**

504 **[A1]**

[Total Marks: 6]

8 **(a)** **(i)** 13 **[B1]**

(ii) 8 or -1 **[B1]**

(iii) $\sqrt{30}$ or π **[B1]**

(iv) -5 or -1 **[B1]**

(b) $\frac{5}{4}$ or $1\frac{1}{4}$ **[B2]**

M1 for $\frac{1}{0.8}$ or $\frac{10}{8}$ or $\frac{8}{10}$ seen.

[Total Marks: 6]

9 **(a)** 36 (or $2^2 \times 3^2$) **[B2]**

Partial working: Identifies 2^2 and 3^2 as common prime factors **[M1]**

(b) $2^3 \times 3^4 \times 5 \times 7$ **[B2]**

Partial working: One incorrect power or missing prime factor **[B1]**

(c) 66 **[B3]**

Partial working: Identifies that C must contain the factor 11 from the LCM **[M1]**

Partial working: Identifies that C must contain exactly $2^1 \times 3^1$ to satisfy the HCF condition **[M1]**

[Total Marks: 7]

10 **(a)** Irrational **[B1]**

(b) Any valid example, e.g. $\sqrt{2}$, $\sqrt{3}$, π (if $y^2 = \pi^2$ which is irrational, so this is wrong)

Only square roots of non-square rational numbers, e.g. $\sqrt{2}$ **[B1]**

(c) **(i)** 53 and Yes **[B2]**

M1 for substituting $n = 3$: $3^2 + 3 + 41$

(ii) Substituting $n = 41$ yields $41^2 + 41 + 41$ **[M1]**

This factorises to $41(41 + 1 + 1) = 41 \times 43$, which has factors other than 1 and itself, therefore it is not prime. **[A1]**

[Total Marks: 6]

11 **(a)** 2 **[B3]**

M2 for $540 = 2^2 \times 3^3 \times 5$ and $180 = 2^2 \times 3^2 \times 5$ written as prime products

or M1 for either 540 or 180 correctly written as a product of prime factors.

(b) $y = 4$ **[B1]**

$z = 3$ **[B1]**

(c) 15 **[B2]**

M1 for identifying the needed powers to reach multiples of 3, e.g. $3^1 \times 5^1$.

[Total Marks: 7]

12 **(a)** 40 **[B2]**

If 0 scored, allow [M1] for setting up the equation $12x = 4 \times 120$ or finding prime factors of 120 indicating $2^3 \times 5$.

(b) **(i)** $2^4 \times 3^2 \times 7$ **[B2]**

If 0 scored, allow [M1] for a correct factor tree or continuous division with at least 2 correct prime division steps.

(ii) 216 **[B3]**

If 0 scored, allow [M1] for writing $72 = 2^3 \times 3^2$.

Allow a further [M1] for deducing that N must contain $2^3 \times 3^3$ to be the smallest perfect cube with the correct HCF (power of 2 is exactly 3, power of 3 is at least 3).

[Total Marks: 7]

13 **(a)** **(i)** pq **[B1]**

(ii) p^3q^3 **[B1]**

(b) $pq = 15$, so the prime numbers are 3 and 5 **[M1]**

$m > n \implies p^3q > pq^3 \implies p^2 > q^2 \implies p > q$ **[M1]**

$p = 5$ and $q = 3$ **[A1]**

(c) $m \times n = (p^3q)(pq^3) = p^4q^4$ **[M1]**

$= (p^2q^2)^2$ which is the square of an integer, hence a square number **[A1]**

(d) Substitutes values: $m = 5^3 \times 3^1$ **[M1]**

Recognises the power of 3 needs to be a multiple of 3, so $k = 3^2 = 9$ **[A1]**

[Total Marks: 9]

14 (a) $p \times q = \text{LCM} \times \text{HCF}$ or $70 \times q = 1260 \times 14$ **[M1]**

$q = \dfrac{17640}{70}$ **[M1]**

252 **[A1]**

(b) Identifying $72 = 2^3 \times 3^2$ or $10080 = 2^5 \times 3^2 \times 5 \times 7$ **[M1]**

Comparing powers of 2 for LCM to establish $x = 5$ **[M1]**

$x = 5$ **[A1]**

$y = 2$ **[A1]**

(c) Factorising expression to $n(n + 1)$ **[M1]**

Valid statement explaining that n and $n + 1$ are consecutive integers, therefore one must be a multiple of 2 (even), making their product always even **[A1]**

[Total Marks: 9]

15 $x = 2$ **[B2]**

$y = 3$ **[B2]**

If 0 scored, award M1 for expressing $180 = 2^2 \times 3^2 \times 5$ or $12600 = 2^3 \times 3^2 \times 5^2 \times 7$

[Total Marks: 4]

16 (a) 36 (or $2^2 \times 3^2$) **[B1]**

(b) $2^3 \times 3^3 \times 5 \times 7$ **[B2]**

If 0 scored, allow [M1] for a product with at least three correct prime bases and powers, or for the correct evaluation 7560.

(c) 10 **[B2]**

If 0 scored, allow [M1] for identifying required missing powers 2×5 or $2^1 \times 5^1$.

[Total Marks: 5]

17 (a) Identifies HCF as $2^1 \times 3^1$ **[M1]**

$x = 1$ **[A1]**

(b) Prime factorises $8100 = 81 \times 100 = 3^4 \times 2^2 \times 5^2$ **[M1]**

Compares highest powers of 3 between A (3^3) and C (3^y) to the LCM (3^4) **[M1]**

$y = 4$ **[A1]**

(c) $\sqrt{11}$ **[B1]**

4π **[B1]**

[Total Marks: 7]

18 (a) (i) Any correct rational number (e.g., 0.45 or $\frac{9}{20}$) **[B1]**

(ii) Any correct irrational number (e.g., $\sqrt{17}$ or $\pi + 1$) **[B1]**

(b) Two different prime numbers identified **[M1]**

Correct sum equating to a square number (e.g., $2 + 7 = 9$ or $13 + 3 = 16$) **[A1]**

(c) 21 **[B1]**

(d) 12 or a^2b^2 seen in the final answer **[M1]**

$12a^2b^2$ **[A1]**

[Total Marks: 7]

19 (a) 12 **[B2]**

If 0 scored, M1 for $84 = 2^2 \times 3 \times 7$ and $120 = 2^3 \times 3 \times 5$ OR for a common factor of 4 or 6 identified **[M1]**

(b) 840 **[B2]**

If 0 scored, M1 for $2^3 \times 3 \times 5 \times 7$ **[M1]**

(c) $\frac{5}{4}$ or $1\frac{1}{4}$ **[B2]**

If 0 scored, M1 for $\frac{1}{0.8}$ or $\frac{10}{8}$ seen **[M1]**

(d) Irrational, because 120 is not a perfect square (or $\sqrt{120}$ cannot be written as a simple fraction) **[B2]**

If 0 scored, B1 for Irrational with no/incorrect reason **[B1]**

[Total Marks: 8]

20 (a) (i) $x = 2$, $y = 8$, $z = 10$ **[B2]**

Partial working: Lists correctly prime numbers $x < 10$ and cube numbers $y > 1$ **[M1]**

(ii) $x = 7$, $y = 8$, $z = 15$ OR $x = 2$, $y = 64$, $z = 66$ **[B2]**

Partial working: Identifies another valid combination of a prime and cube summing to a triangle number **[M1]**

(b) (i) e.g. $p = \sqrt{2}$, $q = -\sqrt{2}$ OR $p = 1 + \sqrt{3}$, $q = 1 - \sqrt{3}$ **[B1]**

(ii) e.g. $p = \sqrt{2}$, $q = \sqrt{8}$ OR $p = \sqrt{3}$, $q = 2\sqrt{3}$ **[B1]**

[Total Marks: 6]

21 **(a)** $x = 2$ **[B1]**

$y = 2$ **[B1]**

M1 for $84 = 2^2 \times 3 \times 7$ OR $3528 = 2^3 \times 3^2 \times 7^2$ seen **[M1]**

(b) 1764 **[B1]**

(c) 10:21:00 **[B3]**

M2 for LCM of 45 and 72 = 360 seconds (or 6 minutes) **[M2]**

If M2 not scored, M1 for listing multiples of 45 and 72 with at least one common multiple, OR $45 = 3^2 \times 5$ and $72 = 2^3 \times 3^2$ **[M1]**

[Total Marks: 7]